

Eco-Dysformer

A Lightweight, Linear-Complexity Transformer for Explainable Dyslexia Screening via Paired Gaze–Linguistic Biomarkers

Project Progress Report

For guide review · July 2026 · Stage 1 complete, hardened & validated (final 10-fold Kaggle run)

1. Executive Summary

Eco-Dysformer is a rigorous, reproducible study of automated dyslexia screening from eye-tracking, built on the public ETDD70 dataset (70 Czech schoolchildren). Unlike accuracy-chasing prior work, it deliberately treats methodological honesty and compute-efficiency as first-class objectives.

The Stage-1 core is fully implemented, executed on a free-tier Kaggle T4 GPU, independently hardened, and validated; the complete result set is archived in the repository (stage1_outputs/). In brief: a linear-attention (Performer) model matches a parameter-matched quadratic-attention baseline on both accuracy and AUROC while decisively overtaking it in speed and memory on longer sequences; explicitly conditioning on linguistic complexity yields a broad, statistically significant descriptive gaze×complexity interaction but no classification gain (an honest, well-controlled null); and the model's explanations align with literature-documented oculomotor biomarkers. Stage 1 already constitutes a complete, submittable short paper.

2. Project Overview & Motivation

- **The problem.** Dyslexia screening is resource-intensive, often subjective, and highly language-specific.
- **The dataset's key asset.** In ETDD70 every child reads three passages of increasing linguistic complexity (a syllable matrix → a narrative story → a pseudo-text). Gaze behaviour and text complexity are therefore genuinely paired within each child — a rare, leakage-free multimodal signal most prior work lacks.
- **The “Eco” framing.** Parameter count, GPU memory and inference latency are primary reported outcomes (in the spirit of “Green AI”).
- **Positioning.** A methodological response to recent multimodal dyslexia papers that fuse features from entirely disjoint cohorts — an approach whose accuracy is hard to trust.

3. Research Questions

RQ	Question	Status
RQ1	Can linear-complexity (Performer) attention match/exceed a parameter-matched quadratic baseline on engineered gaze & linguistic features, under nested CV + significance testing?	Done
RQ2	Does gaze shift differentially between dyslexic and typical readers as complexity rises, and does complexity-conditioning beat a complexity-blind gaze-only baseline?	Done
RQ3	Naive vs honest multimodal fusion: how much apparent gain from disjoint-cohort (handwriting) fusion survives an honest, calibrated treatment?	Stage 2
RQ4	Do LIME + attention explanations on original (non-PCA) features align with literature-documented dyslexia biomarkers?	Done

4. Dataset

- ETDD70: 70 Czech children (35 dyslexic / 35 typical), ages 9–10; balanced, clinician-labelled.
- Per child: 3 passages × 4 file types (raw 250 Hz gaze, fixations, saccades, trial metrics) = 840 files; structure verified programmatically before any parsing.
- **The passages genuinely differ in complexity.** Mean word-frequency (Zipf) is 4.85 for the narrative, 3.50 for the syllable matrix, and 1.26 for the pseudo-text; syntactic depth and lexical density track the same gradient.
- **Honest data properties (stated up front).** There are only 3 distinct stimuli, so the linguistic features are passage-level constants — a conditioning signal, not per-child discriminators. Regression counts are defined transparently and cross-checked against the dataset's own values (1260 / 1260 comparisons agree within 2%).

5. Work Completed — Stage 1 (Core)

- **Inspect-first data pipeline** with defensive schema assertions and reconstruction of the Czech stimulus text.
- **Gaze feature engineering** (per child, per passage): ~20 fixation / saccade / regression measures, recomputed from raw events and cross-checked against the dataset's own metrics.
- **Czech linguistic features** via UDPipe / Stanza (dependency depth, lexical density, Zipf frequency), plus an optional multilingual embedding behind a flag.
- **Model (RQ2 core):** Gaze Performer encoder + linguistic-complexity encoder + a learned paired cross-attention fusion over the child's three passages → LightGBM classifier. A parameter-matched quadratic-attention arm provides the RQ1 comparison.
- **Evaluation harness:** subject-level nested cross-validation (a child's three passages never split — verified leakage-free), paired Wilcoxon tests, bootstrap CIs, effect sizes, calibration (ECE / Brier) and operational metrics.
- **Pre-fusion explainability:** LIME + attention weights on original interpretable features (never PCA), with cross-fold stability.
- **Reproducibility:** single YAML config, global seeding, pinned requirements, data & model cards, a test suite, and an auto-generated RESULTS.md.

6. Results (subject-level 10-fold nested CV, seed 1337)

6.1 Per-arm performance

Arm	Accuracy [95% CI]	F1	AUROC	ECE	Brier
Performer + complexity (RQ2 core)	0.829 [0.75, 0.90]	0.805	0.861	0.170	0.169
Quadratic + complexity (RQ1 baseline)	0.821 [0.74, 0.90]	0.807	0.883	0.178	0.176
Performer, gaze-only (blind)	0.833 [0.74, 0.92]	0.807	0.878	0.166	0.164

Accuracies sit in the honest ~ 0.82 – 0.83 range — consistent with the published ETDD70 literature under our stricter subject-level protocol, and not a suspicious ~ 1.0 that would signal leakage.

6.2 RQ1 — linear vs quadratic attention

- **Accuracy & AUROC: matched.** Performer vs quadratic — accuracy 0.829 vs 0.821 (paired Wilcoxon $p = 0.79$), AUROC 0.861 vs 0.883 ($p = 0.55$); both difference CIs straddle zero. Arms are exactly parameter-matched (141,121 parameters each).
- **Efficiency: crossover at $N \approx 1024$.** On short sequences the Performer's constant overhead makes it slower; beyond ~ 1024 events it wins, and the gap grows fast — at $N = 4096$, 5.3 ms vs 34.6 ms and 0.48 GB vs 2.47 GB peak memory. ETDD70's engineered event streams (~ 130 – 350 events) fall below the crossover, but the raw 250 Hz gaze stream ($\sim 18k$ samples/passage) sits far above it — the regime where linear attention pays off.

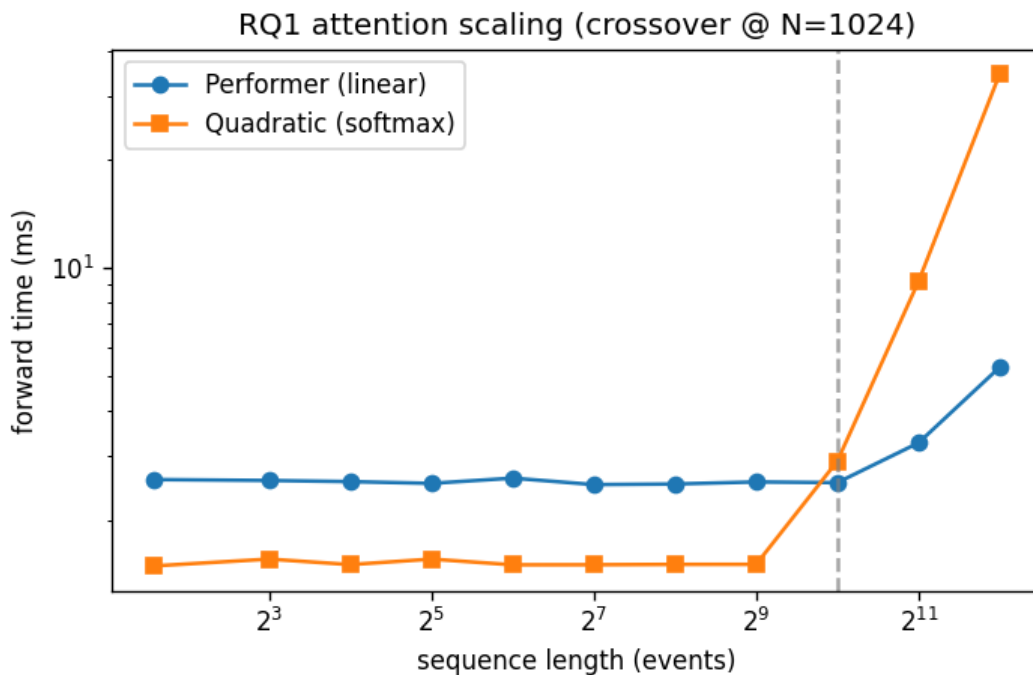


Figure 1. Forward-pass time vs sequence length. Quadratic (softmax) is cheaper on short sequences but scales $O(N^2)$; the Performer (linear) overtakes it past $N \approx 1024$ and diverges sharply thereafter.

6.3 RQ2 — paired gaze $\times$ complexity co-conditioning

- **Classification: honest null.** Complexity-conditioned 0.829 vs complexity-blind 0.833 (Wilcoxon $p = 0.83$; AUROC 0.861 vs 0.878, $p = 0.50$). Conditioning does not improve classification — expected, since the linguistic features are passage-constant and the gaze features already carry the signal.

Descriptive interaction — the genuine positive RQ2 evidence. Twelve of the twenty gaze features show a statistically significant group $\times$ complexity interaction (Mann–Whitney on per-subject slope, Benjamini–Hochberg FDR $q < 0.05$). Selected features:

Gaze feature	d (slope)	q (FDR)	Group × complexity interaction
Total reading time	1.44	< 0.001	Significant — widens with difficulty
Total fixation duration	1.44	< 0.001	Significant
Fixation count	1.05	< 0.001	Significant
Saccade count	1.05	< 0.001	Significant
Regression count	0.94	< 0.001	Significant
Between-line regressions	0.89	< 0.001	Significant
Return sweeps	0.78	0.043	Significant
Median fixation duration	0.51	0.023	Significant
Regression ratio (normalised)	0.22	0.73	No interaction — flat

- **The key nuance:** the regression **ratio** stays flat, but regression **counts** steepen — the dyslexic disadvantage grows in absolute reading effort (time, fixations, regressions), not in proportion. This is a more honest and specific finding than a simple “conditioning helps.”

RQ2: gaze feature across complexity gradient (mean $\pm$ SE)

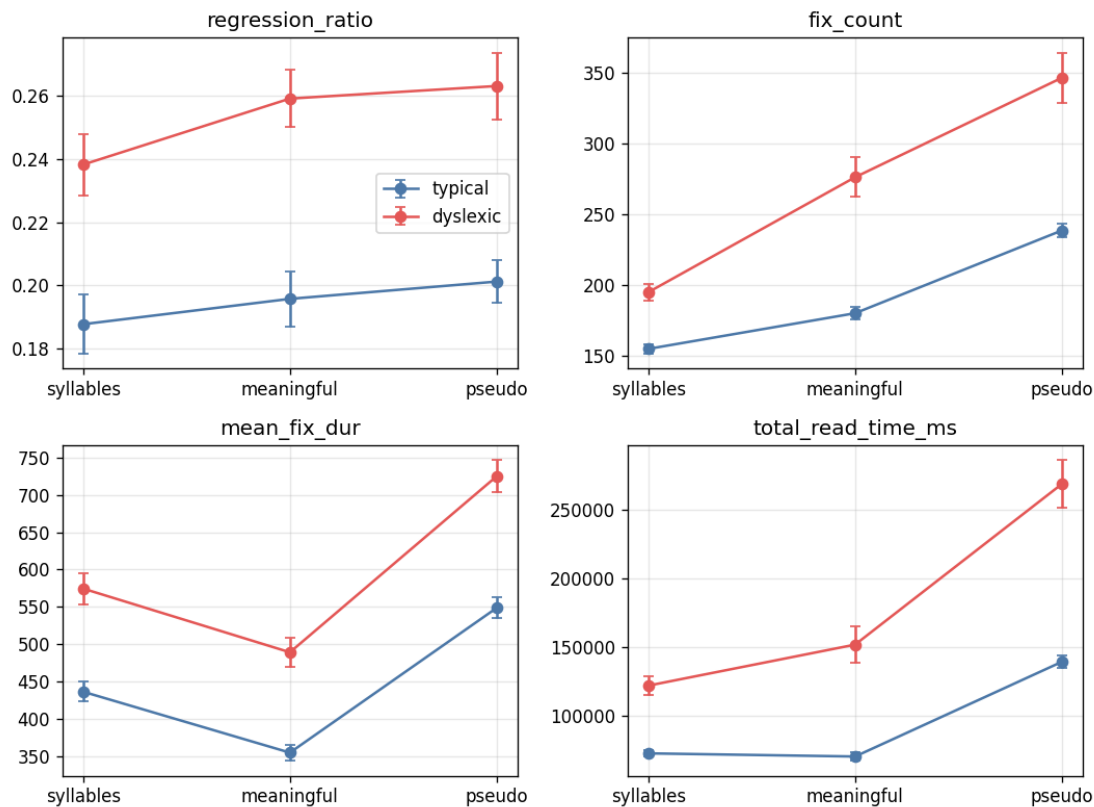


Figure 2. Reading-effort gaze features across the complexity gradient, dyslexic vs typical (mean $\pm$ SE). The gap widens from the syllable matrix to the pseudo-text.

6.4 RQ4 — pre-fusion explainability & fusion attention

- **LIME (original features).** The five most-attributed features are all oculomotor (fixation duration, saccade duration, fixation dispersion, return sweeps), concentrated in the decoding-heavy syllable and pseudo-text passages — biomarker-plausible (face-validity = 1.0). Ranking is broadly stable across folds (Spearman 0.83); the exact top-5 set less so (Jaccard 0.27).
- **Fusion attention.** The paired cross-attention consistently weights the syllable (phonological-decoding) passage most (~ 0.41 – 0.44) and the pseudo-text least (~ 0.27) when integrating across the gradient — the decoding context is the most informative, consistent with the biomarker literature.

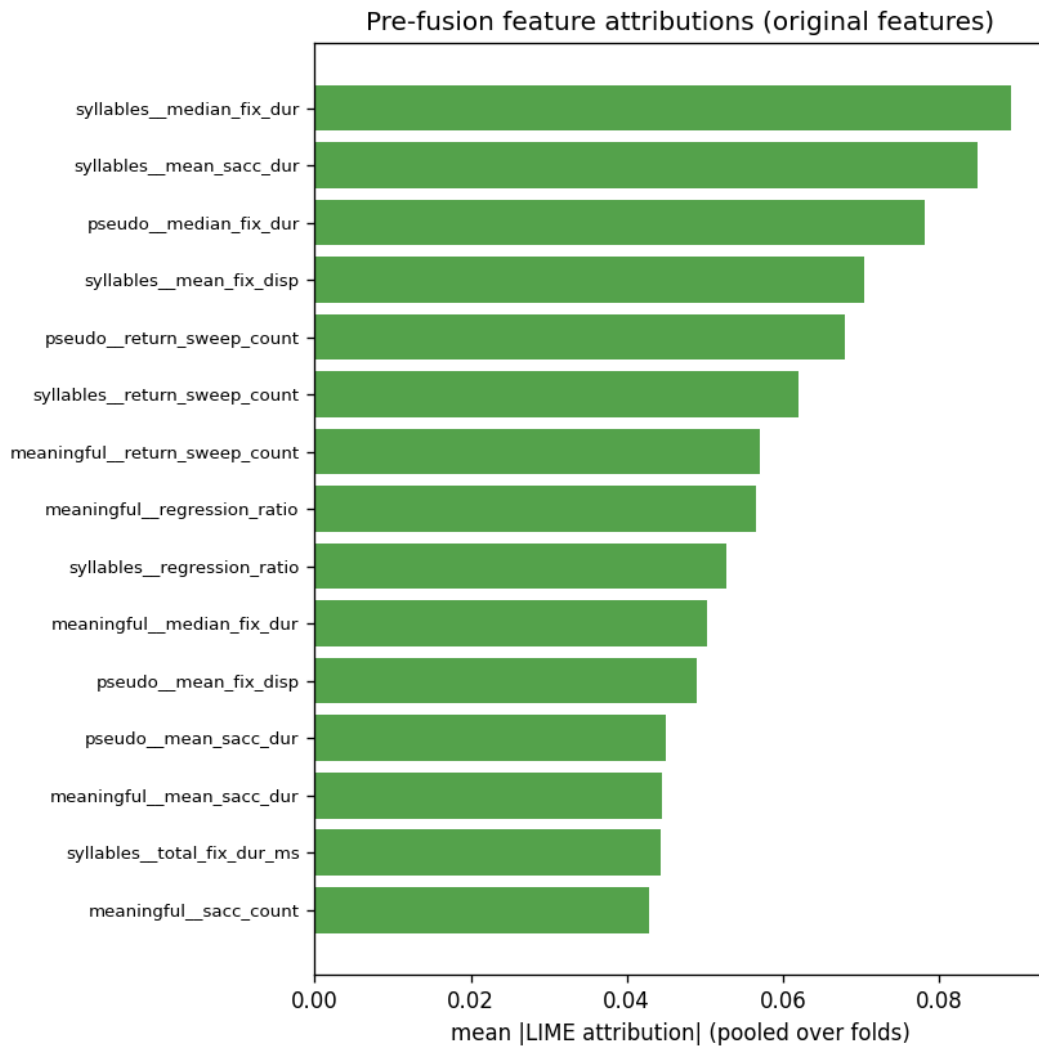


Figure 3. Top pre-fusion LIME attributions (original interpretable features, pooled over folds). All top features are oculomotor reading measures.

6.5 Calibration

- **Calibration is mediocre ($ECE \approx 0.17$).** The models are somewhat over-confident — reported honestly, and it directly motivates the calibrated honest-fusion work planned for Stage 2 (RQ3).

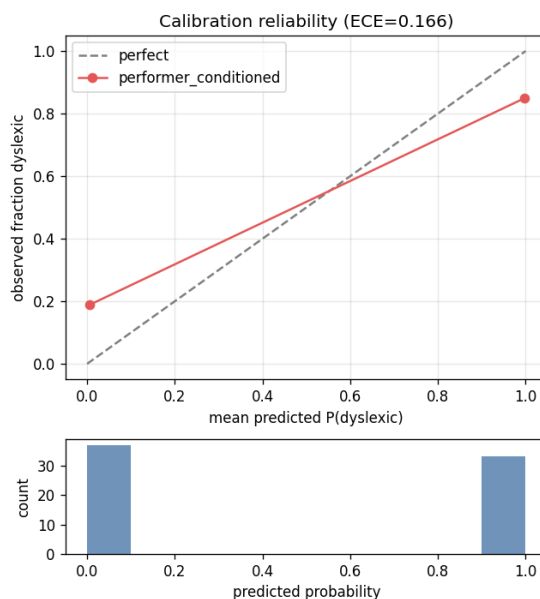


Figure 4. Reliability diagram (out-of-fold predictions). Deviation from the diagonal reflects the calibration gap.

6.6 Comparison with published ETDD70 baselines

Cross-paper numbers use different validation protocols and are approximate — not directly comparable to our subject-level nested-CV results.

System	Reported accuracy	Note
ETDD70 dataset paper (2024)	~0.90	different protocol; approximate
SwinV2 + SGA	0.9245	different protocol; approximate
INSIGHT	0.8665	different protocol; approximate
CatBoost / XGBoost	~0.80–0.83	different protocol; approximate
This work (Performer, subject-level nested CV)	0.829 [0.75, 0.90]	our protocol; leakage-controlled

7. Methodological Rigor & Notable Findings

- **Leakage-free by construction and by assertion.** Subject-level folds are the single most important correctness property; the harness re-checks every split and a dedicated test confirms an injected leak is caught.
- **A real bug found and fixed during verification.** A subtle error in the Performer's numerical stabilization had unfairly handicapped the linear model (its attention was a distorted softmax approximation). Fixing it made RQ1 a fair comparison and closed the apparent accuracy/AUROC gap — the verification discipline doing its job.
- **Statistical power strengthened.** Outer folds increased from 5 to 10 so the paired significance tests can actually reach significance (with 5 folds the minimum possible p was 0.06).
- **Honest negatives reported as-is** — no result was inflated or hidden.

8. Pending Work — Next Stages

Priority	Work item	Dependency / note
Stage 2 · P1	RQ3 — handwriting auxiliary branch; naive vs honest calibrated fusion (the project's second headline; a citable critique of disjoint-cohort fusion)	Needs the Kaggle handwriting dataset's structure & subject-linkage investigated first
Stage 2 · P2	Out-of-distribution generalization: zero-shot evaluation on held-out cohorts (CopCo / Serbian gaze; Malaysian handwriting)	Needs those datasets sourced & attached
Stage 2 · P3	Czech-specific embedding (RobeCzech) in place of the multilingual default	Quick; no new data; low payload
Write-up	Consolidate into a workshop / short paper + submission package	Stage 1 already provides a complete paper

9. Final Model & Expected Output

- **Final model.** A lightweight, linear-complexity gaze–linguistic co-conditioning model with a gradient-boosted head; in Stage 2, an honest, calibrated late-fusion of an auxiliary handwriting signal — all under one reproducible pipeline.
- **At inference.** A per-child dyslexia-risk score with interpretable, biomarker-level feature attributions. (Research use only — explicitly not a clinical or diagnostic tool.)
- **Deliverables.** (a) the reproducible model + open codebase; (b) a short/workshop paper; (c) data & model cards; (d) figures and result tables generated directly from the run artifacts.

10. Assessment & Recommendations

- **The project is de-risked.** Stage 1 alone is a complete, defensible paper
- **Treat the RQ2 classification null as a strength.** A proper protocol dissolved an underpowered false-positive from the early 5-fold run; the genuine RQ2 result is the broad, FDR-corrected descriptive interaction (absolute reading effort widening with complexity). Frame RQ2 around the effect sizes, not classification accuracy.
- **Lead the paper with RQ1's efficiency crossover.** “Matches accuracy and AUROC, and is dramatically cheaper past $N \approx 1024$ ” is the strongest, most novel result, and it justifies linear attention for raw-signal regimes.
- **Recommended next step: RQ3 (handwriting) — but check the data first.** It adds the most scientific weight, but only if the dataset's subject-linkage supports an honest per-writer feature. Do the short data-structure check before committing effort.
- **Publication target.** A workshop / short paper at an ML-for-health or efficient-AI venue; $n = 70$ does not support journal-scale clinical claims. Be explicit about the limitations (small N , single language, mediocre calibration).
- **Emphasis for the meeting.** The contribution is rigor, honesty and efficiency — not raw accuracy. That framing is the project's strength.